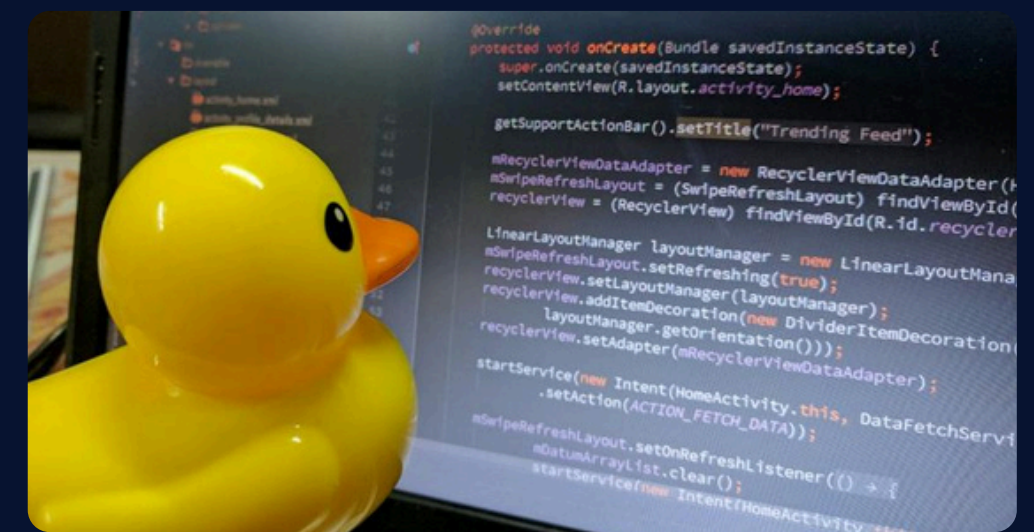




Boosting & Adaboost



Session -> 11



Intro to Ensemble Learning

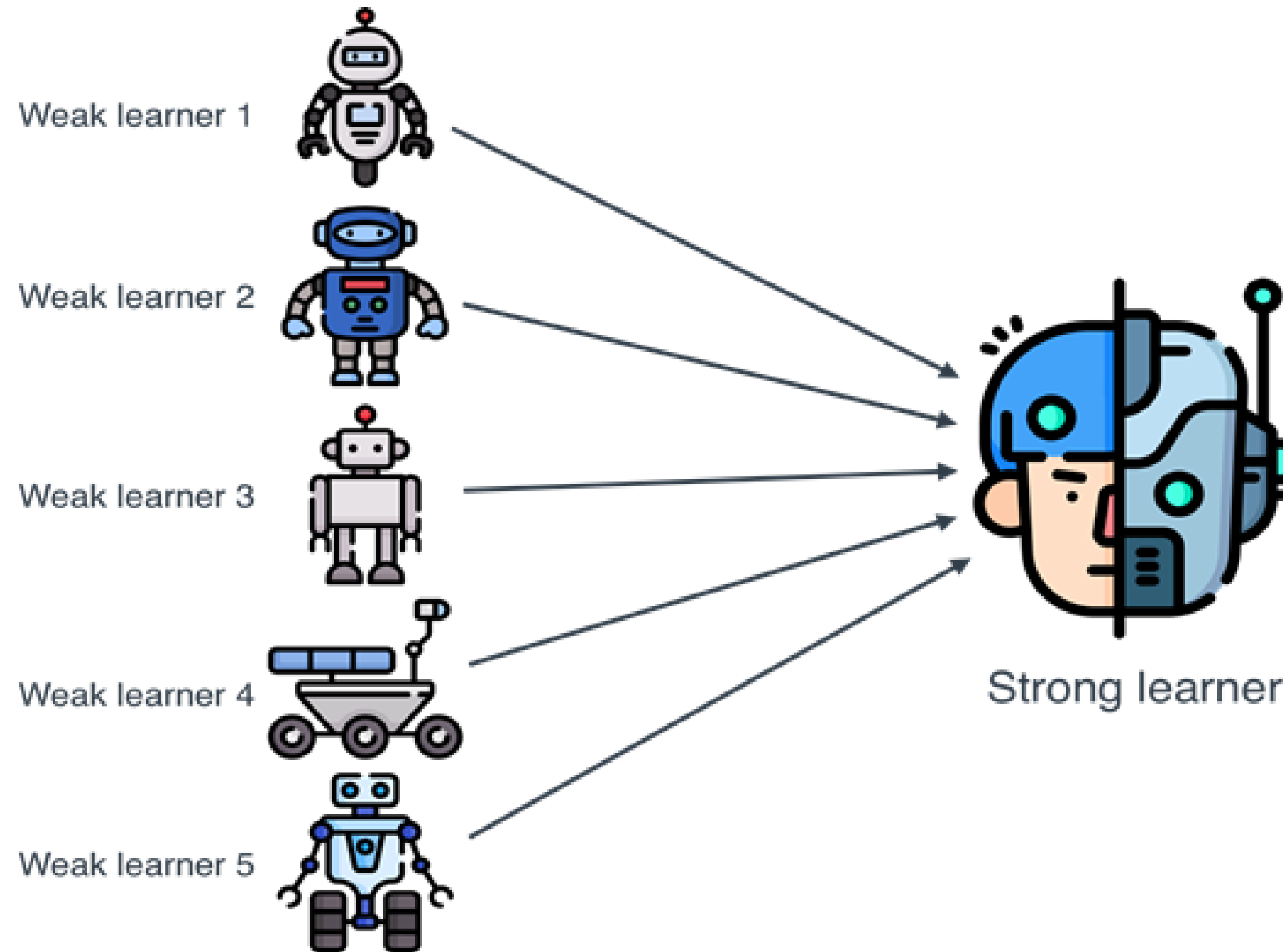
What is Ensemble Learning?

- A machine learning technique that combines multiple models (often called base learners or weak learners) to solve a problem and improve overall performance compared to individual models.

The methods that we will cover in this course:

- Bagging
- Boosting

What is Ensemble Learning?



Boosting

- Boosting is an ensemble method that trains multiple models sequentially, where each new model focuses on correcting the mistakes made by the previous ones. Unl
- Boosting is similar to bagging in building strong model from weak learners, But Unlike bagging, **boosting does not use random sampling** – it learns from the entire dataset but gives more weight to difficult examples.

Boosting

- For classification tasks, it combines the predictions of all weak learners using a **weighted vote**, where more accurate models have more influence.
- For regression tasks, it computes a **weighted average** of the predictions, with more accurate models contributing more to the final output.

Why Boosting is useful?

- Improves Accuracy.
- Reduces Bias and Variance: Boosting reduces both bias (underfitting) and variance (overfitting), making models more robust.
- Handles Complex Data Patterns: Boosting can capture complex relationships in data by iteratively focusing on hard-to-predict examples.

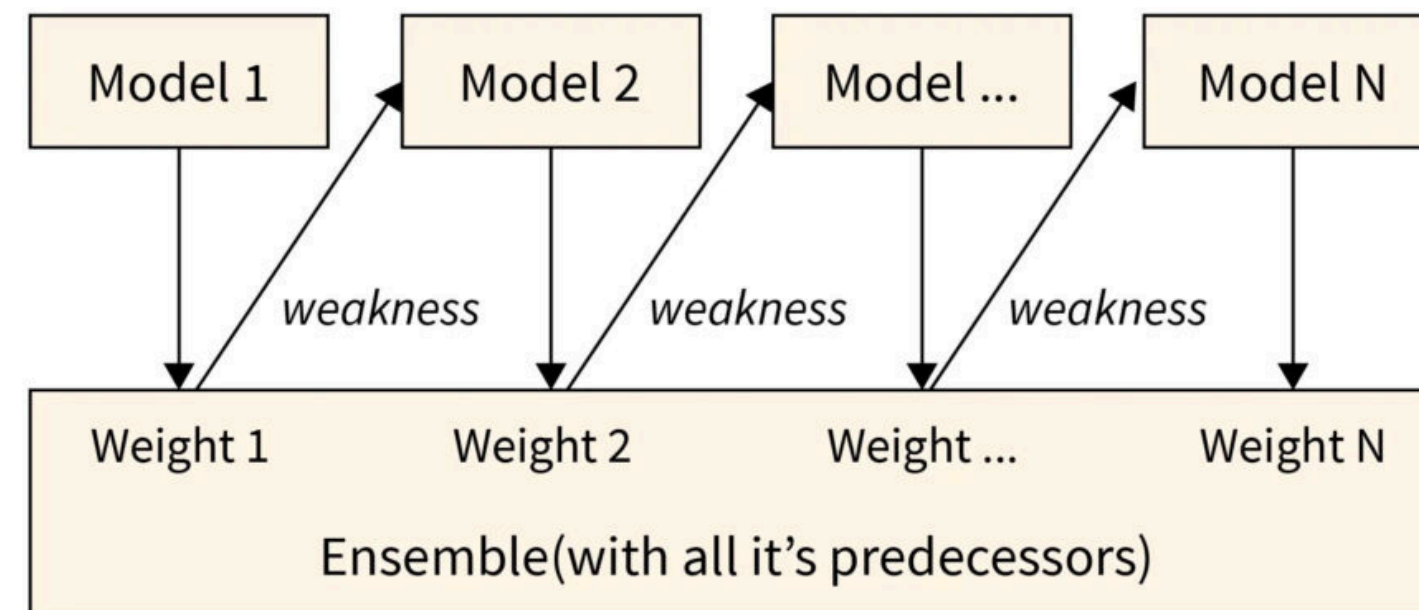


AdaBoost

What is AdaBoost?

- AdaBoost (Adaptive Boosting) is an ensemble learning method that combines multiple weak learners to create a stronger model. AdaBoost trains these weak learners in sequence, focusing on data points that were misclassified in previous rounds.

Model 1,2,...,N are individual models (eg. decision tree)



How AdaBoost Works?

- 1- Train the first weak learner on the dataset.
 - Each data point is initially given equal weight.
- 2- After a weak learner is trained, the points are modified as follows:
 - Points that are misclassified are given more weight.
 - Points that are correctly classified are given less weight.
- 3- Train a new weak learner on the updated (weighted) dataset.

How AdaBoost Works?

4- Calculate the importance of each learner based on its accuracy using that formula:

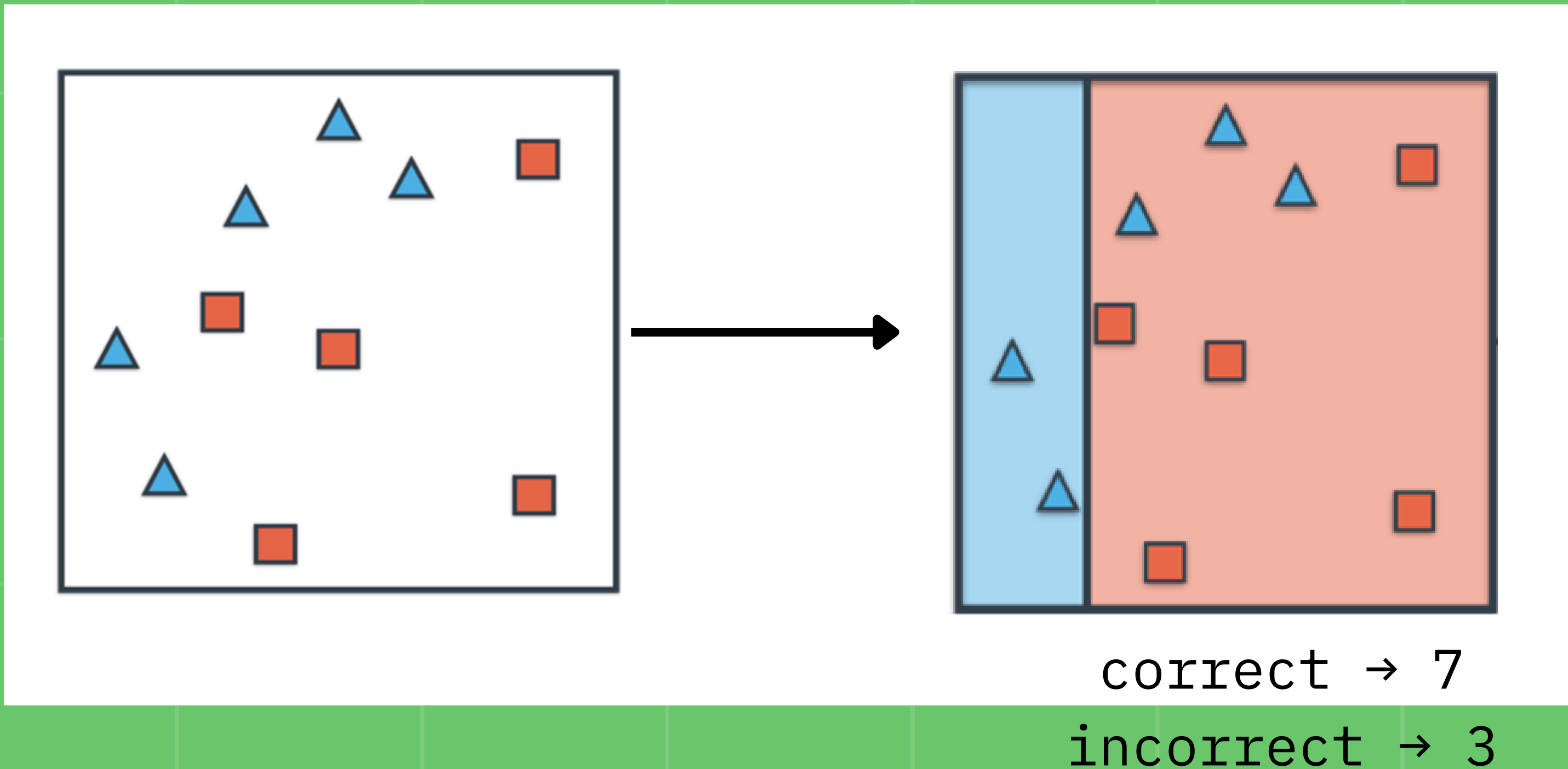
$$\ln (\text{accuracy} / 1 - \text{accuracy})$$

5- Repeat Steps 2 to 4 for a predefined number of rounds (T) or until the error is low.

6- Combine each model's predictions weighted by its importance.

AdaBoost Example

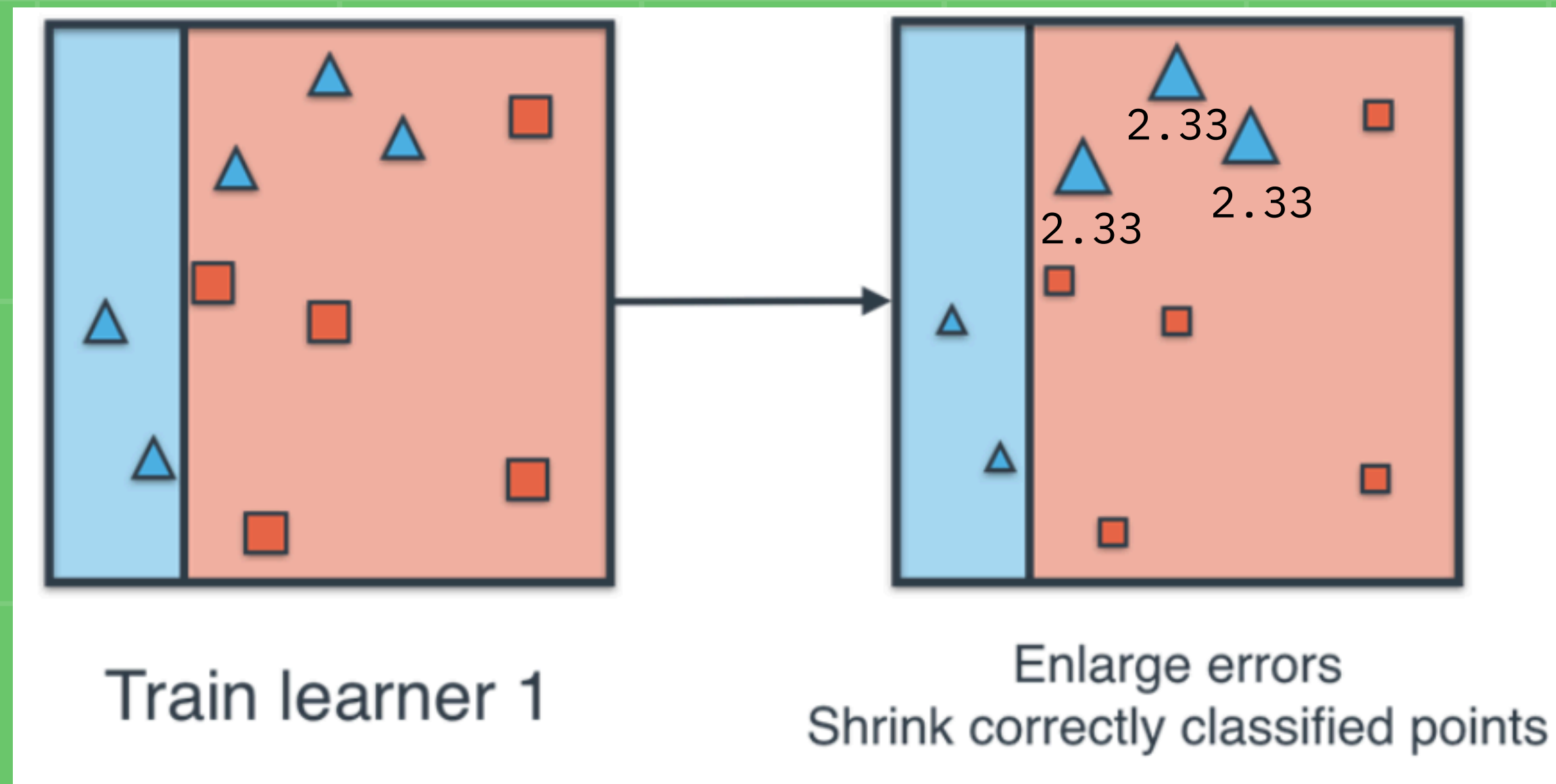
Step 1: Fit the first weak learner to the dataset



AdaBoost Example

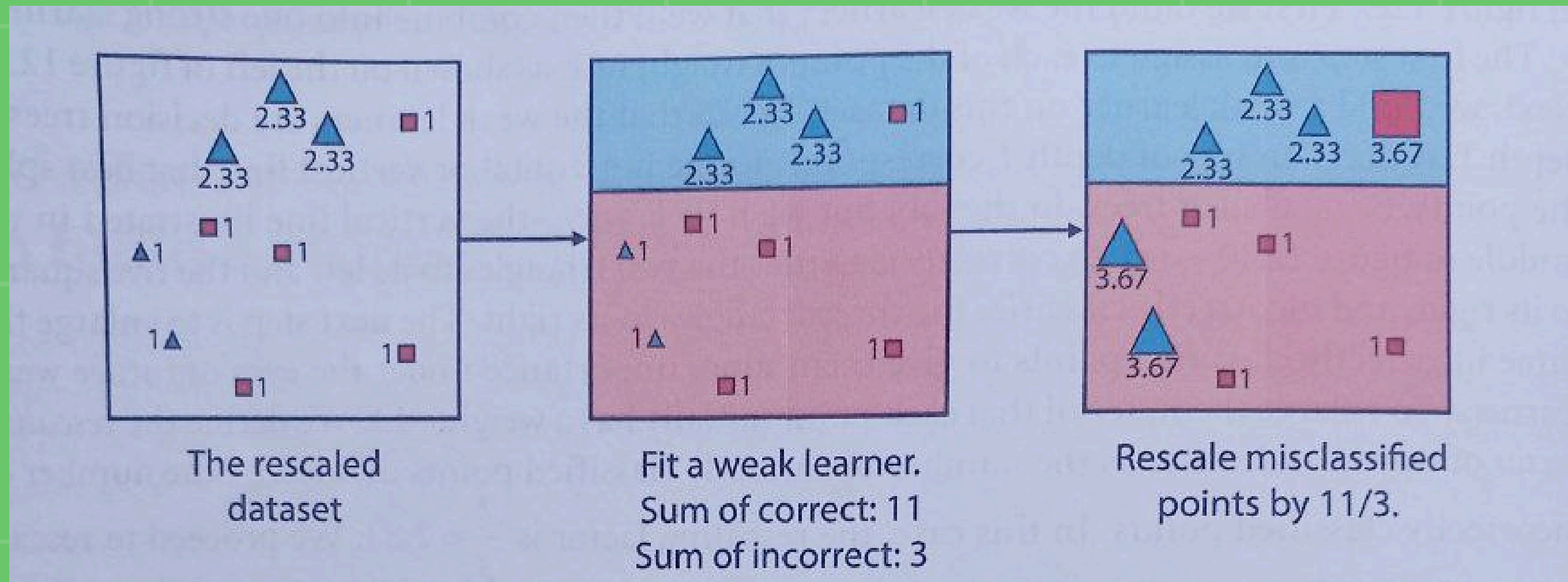
Step 2: Enlarge the weight of misclassified points

$$\text{weight} = \text{correct} / \text{incorrect} = 7/3 = 2.33$$



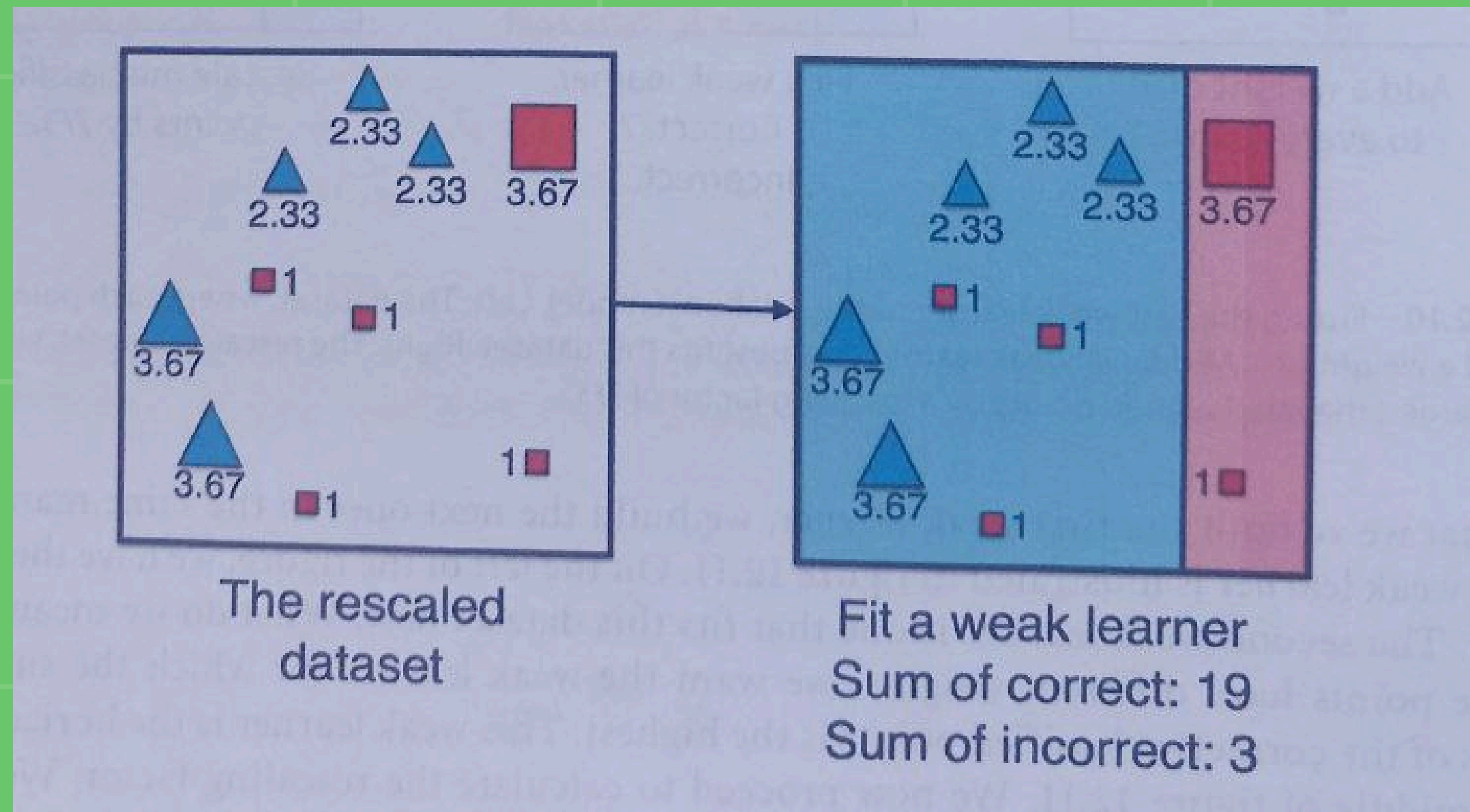
AdaBoost Example

Step 3: Keep training weak learners sequentially and enlarge errors



AdaBoost Example

Step 3: Keep training weak learners sequentially and enlarge errors



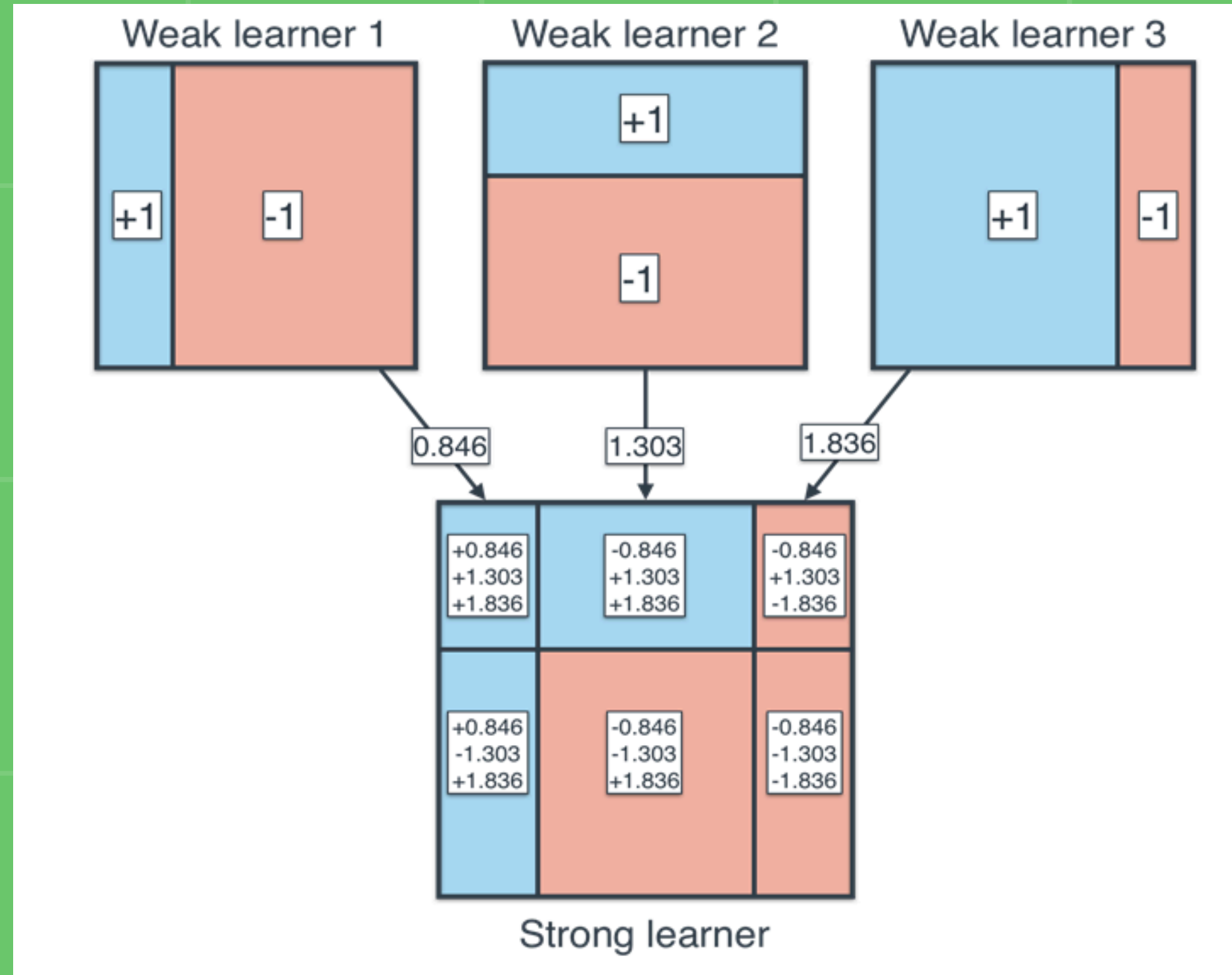
AdaBoost Example

Step 4: Calculate the importance of each model

- Weak Learner (1): accuracy = $7/10$
importance score = **0.847**
- Weak Learner (2): accuracy = $11/14$
importance score = **1.299**
- Weak Learner (2): accuracy = $19/22$
importance score = **1.846**

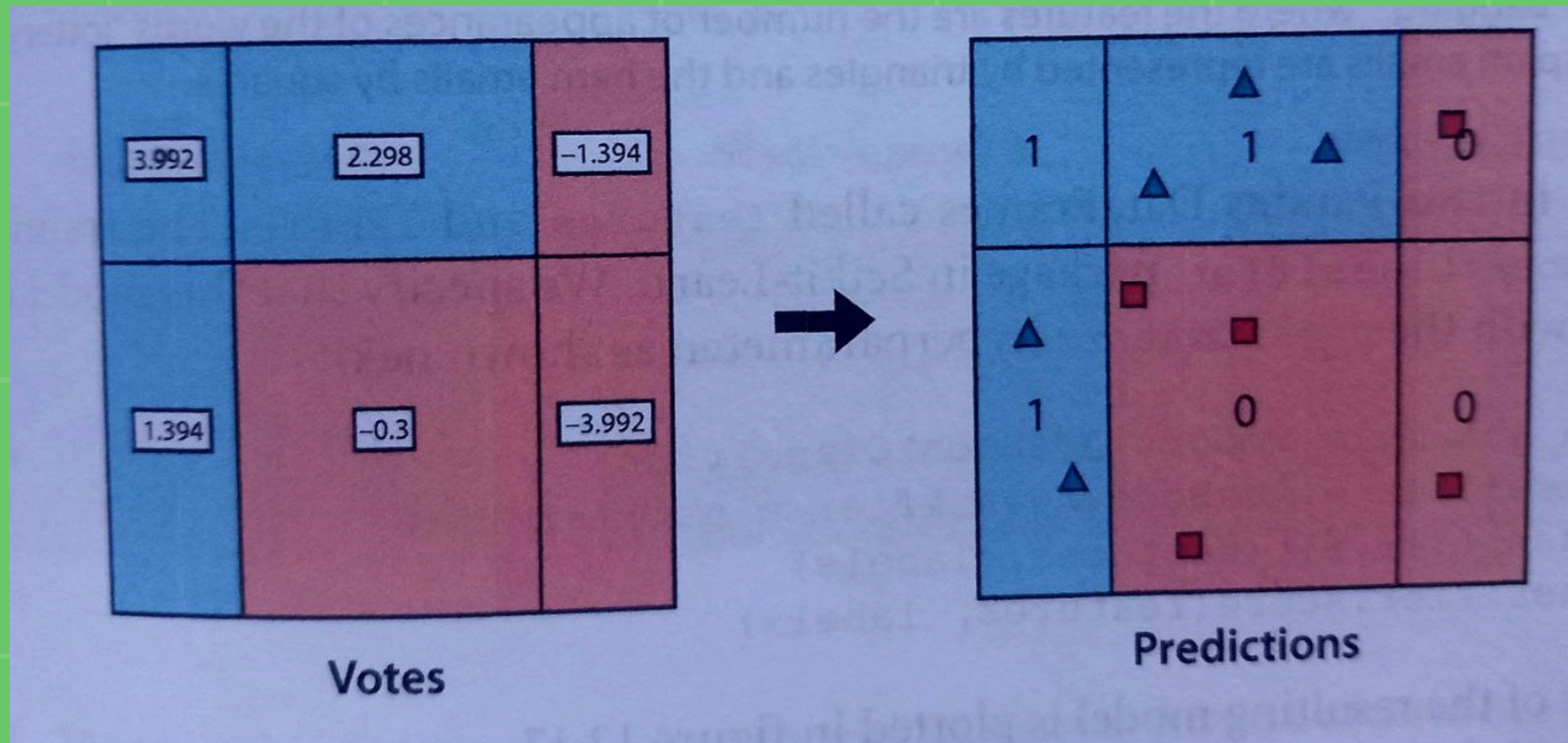
AdaBoost Example

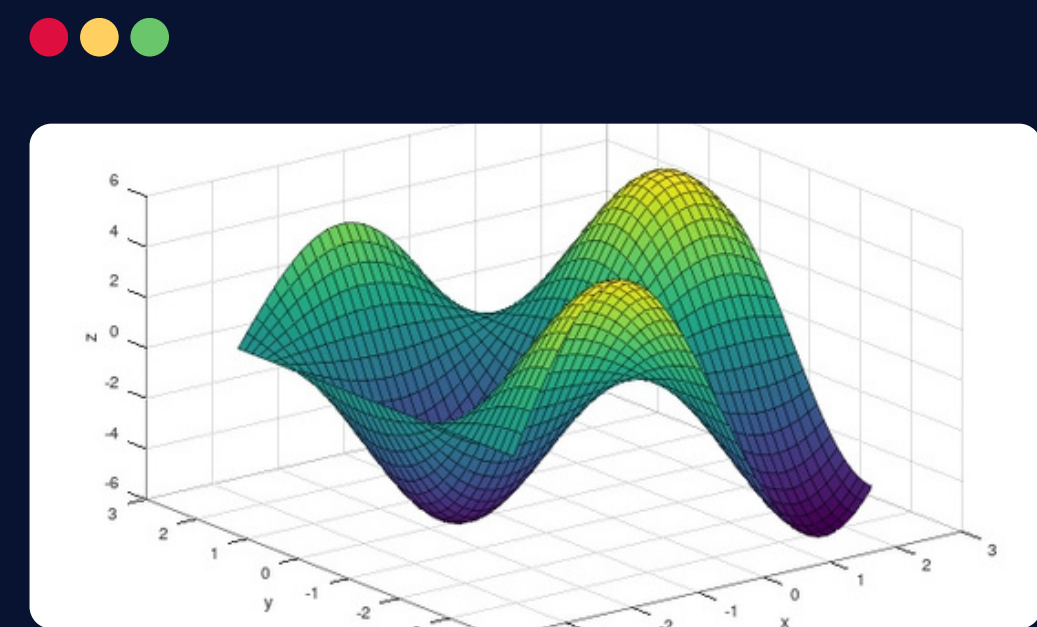
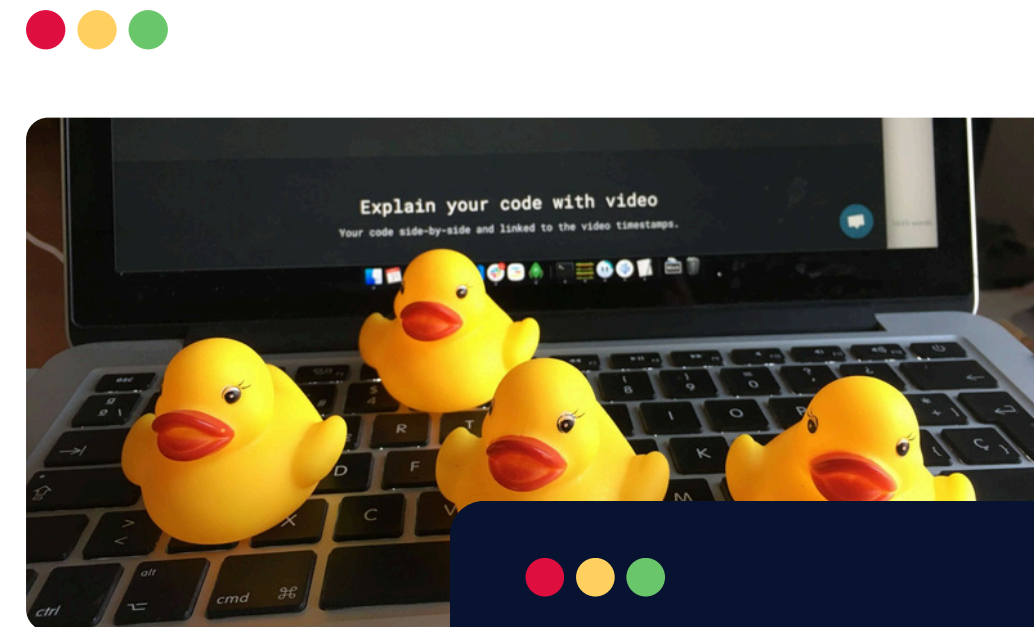
Step 5: Combine the models predictions using weights (importance score)



AdaBoost Example

Step 5: Combine the models predictions using weights (importance score)





Thank You

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